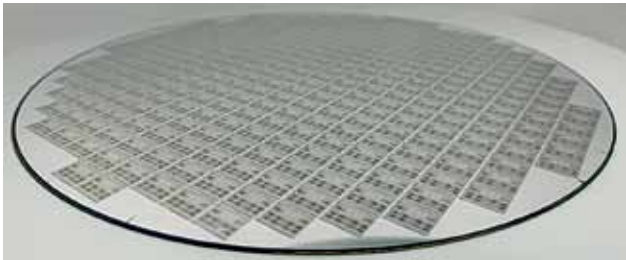




Memristor-based chip promises faster RF signal processing

Researchers from UMass Amherst, Texas A&M, and TetraMem (New England) have developed a novel SoC that uses memristors to process analogue RF signals faster

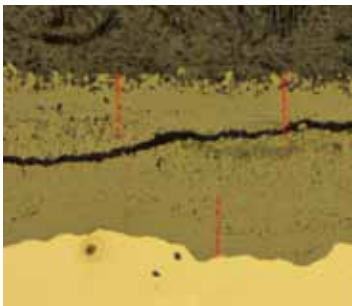


A 300mm wafer with many memristive systems-on-a-chip
(Credit: Yi Huang)

and more efficiently than traditional SDRs. Mimicking the brain's real-time sensory processing, the chip eliminates the need for ADCs, reducing energy and latency. With AI inference and RF processing spread across ten cores, it is a key enabler for future 6G and advanced Wi-Fi systems.

AI-driven method detects and assesses corrosion automatically

In a major step toward autonomous industrial monitoring, researchers from the Indian Institute of Science (IISc), Bengaluru, Karnataka, and QSRTC have developed an AI-driven method to detect and assess corrosion in power systems, eliminating the need for human input. Using unsupervised deep learning, the system analyses metal surface images to gauge rust thickness and porosity, key signs of corrosive damage.



Input optical microscopy images marked with deposit thickness measurements
(Image: Ashwin RajKumar)

Achieving 73% accuracy on steam generator tubes, it offers early warnings, cuts inspection time, and promises safer, low-maintenance, digitally transformed operations.

New standard strengthens V2X systems

Keysight Technologies (UK) and Korea's Automotive Technology Institute (KATECH) have co-developed a new global testing standard, ISO/TR 17716:2025, to enhance the safety and reliability of connected vehicles. This standard

closes existing gaps in V2X radiated immunity testing by evaluating communication performance under electromagnetic interference (EMI). Combining Keysight's EMC testing

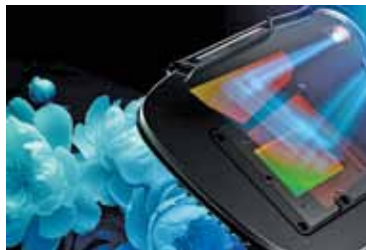


Keysight and Korea Automotive Technology Institute team up to improve vehicle-to-everything component testing

expertise with KATECH's V2X know-how, the validated procedures ensure uninterrupted, safe vehicle-to-infrastructure communication critical for autonomous driving in EMI-heavy environments.

3mm-thick holographic display that delivers lifelike 3D visuals

Stanford researchers (California) have unveiled a 3mm-thick holographic display that delivers lifelike 3D visuals using true holography, not stereoscopy. Combining a



Holographic virtual reality displays enable lifelike 3D experiences in ultra-compact devices (Credit: Nathan Matsuda)

custom waveguide, AI calibration, a wide field of view, and a large eyebox, the prototype offers an immersive, lightweight mixed reality experience. This breakthrough boosts comfort and realism, pushing closer to a

visual turing test future where digital blends seamlessly with reality. A commercial version is already in the works.

Better method found to grow large-area, high-quality InSe wafers

Researchers at Peking University (Beijing, China) and Renmin University (Beijing, China) have developed a new 'solid-liquid-solid' method to grow large-area, high-quality indium selenide (InSe) wafers. Their InSe transistors outperform silicon, achieving record room-temperature mobility, sub-10nm scaling, and surpassing 2037 IRDS targets. With uniform 2-inch (5.08cm) wafers and exceptional performance, InSe now emerges as a powerful silicon alternative for AI, smart devices, and beyond.